

A new correlation for single-phase pressure loss through SMV static mixers at high Reynolds numbers

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ABSTRACT

Robust, scalable correlations for pressure loss across corrugated (SMV-style) static mixers at high Reynolds (Re) numbers are currently lacking. To address this, results from an experimental and computational fluid dynamics (CFD) based study involving a single SMV mixing element for Re in the range of 8,000 – 250,000 is reported. The investigated scenarios encompassed three different pipe diameters and variations in the wall roughness. The equivalent sand grain roughness was first estimated from measurements in an empty pipe and provided as an input to the CFD model. The agreement between the CFD predictions and measurements ascertained the validity of the roughness estimate. When the mixing element was absent, roughness contributed about 70–90% of the pressure losses in the rough stainless steel pipes and up to 7% of the losses in the smooth PVC pipe. A capillary model based semi-analytical correlation was developed to generalize the pressure loss characteristics when the mixing element was present. The proposed correlation captured both current and historical pressure loss measurements with an improved accuracy over currently available correlations. Even with the mixing element present, about 60% of the pressure losses were attributed to the inclusion of wall roughness.

1. Introduction

Static mixing technologies have played a substantial role in process intensification efforts over the past century. These fundamentally simple devices harness the kinetic energy of a flowing fluid to effectively mix and blend fluids without the need for stirred vessels or external energy sources. Advantages of low cost, low maintenance, and low power requirements have resulted in static mixers becoming a common tool for both single and multiphase mixing applications [1].

With the recent trend towards process intensification, the overall efficiency and small form-factor associated with static mixing reactors has renewed interest in this area. These types of reactors excel in absorption or fast-reaction scenarios that are constrained by low residence times, and where interfacial diffusion resistance is reasonably low, such as a high pressure, gas-continuous processes [2]. However, since static mixer geometries result in complex flow fields it has been challenging to develop accurate and robust correlations that characterize their performance under dynamic process conditions. It is this lack of knowledge that has led to large amounts of research regarding mixing encompassing both laminar and turbulent regimes. Turunen et al. (1994) studied gas-liquid mass transfer and interfacial area in static mixing reactors [3]. Sanchez and co-workers examined gas-liquid mass transfer and pressure loss in two types of static mixers at low and moderate Reynolds numbers with a gas continuous phase [4]. Jaworski et al. (2007) conducted CFD

modeling of liquid-liquid droplet breakup phenomena in a Kenics style static mixer [5]. Konopacki et al. (2015) used CFD simulation to study a novel static mixer for enhancing heat transfer [6]. Rauline and others examined numerical predictions of pressure loss for several types of static mixing devices in the creeping flow regime [7]. In all studies, understanding the dynamic pressure loss characteristics in static mixing reactors is a critical first-step towards their design, optimization, and scale-up since heterogeneous heat and mass transfer characteristics are known to correlate strongly with pressure loss [8]. At high velocities, when liquid phase resistance to mass transfer is controlling, co-current operation is shown to result in higher mass transfer rates than operating in a counter-current mode [8]. During co-current operation, corrugated mixing elements, like the Sulzer SMV mixer can greatly enhance the turbulent mixing of both liquid-liquid and gas-liquid streams thereby reducing the required pipe lengths for mixing or extraction applications. This corrugated plate type mixer is comprised of channels that split the fluid prior to forcing it back together in cross-flow as shown in Fig. 1.

This same type of contacting scheme has been used for a variety of structured packing options for packed tower systems [8]. This type of a stacked corrugated plate design has been shown to yield high mass transfer rates while minimizing pressure losses and liquid holdup [9,10].

There are three common functional forms employed to quantify the pressure loss (ΔP) across corrugated static mixers:

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Nomenclature

A	Cross-sectional area (m ²)
ΔB	Near-wall velocity distribution factor
C ₀	Empirical model constant
C ₁ , C ₂ , C ₃ , C ₄	Empirical model constant
C _μ	Model constant = 0.09
C _s	Roughness constant = 0.5
C _p	Mixer channel frictional loss coefficient
D	Pipe diameter (m)
D _c	Mixer channel hydraulic diameter (m)
d _H	Mixer hydraulic diameter as defined by Theron et al., 2011 (m)
E	Empirical constant
e	Absolute roughness in Nikuradse equation (m)
f	Fanning friction factor
f _c	Mixer friction factor of Eq. (18)
k	Turbulence kinetic energy (m ² /s ²)
K _s	Dimensional sand-grain roughness (m)
K _s ⁺	Dimensionless equivalent sand-grain roughness
K _s /D	Relative roughness
L	Mixer length (m)
ℓ	Streamline length (m)
$\frac{L}{D}$	Length to diameter ratio of mixing element
P	Pressure (N/m ²)

$\mathcal{P}$	Perimeter of mixer channel (m)
Re = $u_0 D \rho / \mu$	Reynolds number based on D
Re _c = $u_0 D_c \rho \tau / \epsilon \mu$	Reynolds number based on D _c
Re _{cT}	Critical Reynolds number for start of turbulent flow
u ⁺	Dimensionless near-wall velocity
u	Near-wall velocity (m/s)
u ₀	Average pipe flow velocity (m/s)
u _s	Superficial velocity (m/s)
y ⁺	Dimensionless wall distance
y	Dimensional wall distance (m)
Z	Ratio of mixer friction factor to empty-pipe friction factor

Symbols

β	Corrugation angle relative to flow axis (degrees)
Δ	Difference operator
ρ	Fluid density (kg/m ³)
ε	Void fraction of mixing element
μ	Dynamic viscosity (Pa.s)
π	Pi constant
ν _t	Turbulence viscosity (m ² /s)
κ	Von Karmen constant
ν	Kinematic viscosity (m ² /s)
τ _w	Wall shear-stress (kg/m.s ²)
τ	Tortuosity
ζ	Frictional loss coefficient of Streiff et al., 1979

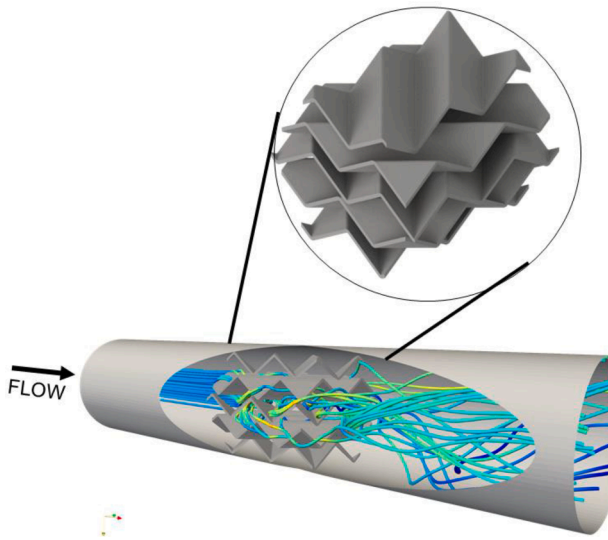


Fig. 1. Flow in a corrugated plate type mixer.

$$Z = \Delta P_{\text{mixer}} / \Delta P_{\text{empty pipe}} \quad (1)$$

The Z-factor [12] in Eq. (1) is defined as the pressure loss across the mixer to that in an empty pipe. Z values generally range from 100 – 200 in the turbulent regime for corrugated plate mixers.

A second method expresses ΔP in terms of the square of the fluid velocity v as given in Eq. (2).

$$\Delta P = C_1 \times v^2 \quad (2)$$

Where C₁ in Eq. (2) is a constant whose value is in the range 105 +/- 5 [11].

The third functional form is a Blasius type correlation where the fanning friction factor (f) is computed as a function of Reynolds number

(Re) as given in Eq. (3).

$$\frac{f}{2} = \frac{\Delta P D}{\rho u_0^2 L} = C_2 + \frac{C_3}{\text{Re}^{C_4}} \quad (3)$$

Where ρ is the velocity of the fluid and appropriate length scales: such as length of the static mixer element (L) and pipe diameter (D) are employed in the calculation. C₂, C₃ and C₄ are constants that are empirically determined. This form has been adopted by several previous studies [8,12].

The application of interest in this study is the absorption of impurities from a high-pressure gas-phase. This situation occurs often in natural gas processing and transport as well as environmental applications. For pipeline of diameter greater than 2", the corrugated-plate mixers have been deemed to be the most suitable choice for absorption [13].

A more sophisticated, semi-analytical model for pressure drop in corrugated mixers was presented and validated for water through 40–80 mm mixer elements by Paglianti et al. (2013) [14]. The model hypothesized that the total pressure drop across a corrugated element is the sum of the friction pressure drop and the interfacial pressure loss due to inlet/outlet effects. The correlation from [14] is expressed as:

$$\Delta P = \frac{\pi d_H L}{A \sin(\beta)} \frac{\zeta \rho u_s^2}{2} + 4 \zeta \frac{L e_{\text{inlet}} + L e_{\text{outlet}}}{d_H} \frac{\rho u_s^2}{2} \quad (4)$$

Where β is the corrugation angle relative to the flow direction and $L e_{\text{inlet}}$ and $L e_{\text{outlet}}$ are the interfacial lengths that account for pressure losses at the interface between multiple elements. The authors did not demonstrate how this value was calculated in practice, and thus the values given in the paper were assumed to be generally applicable. The term ζ is the effective friction factor of the materials and was taken from previous work by the authors [13].

Building on prior work [15], a model was presented which accounts for four separate cases: (1) a single element with a L/D > 1; (2) multiple elements with L/D > 1; (3) a single element with L/D < 0.56; (4) multiple elements with L/D < 0.56. It was noted that fluid bypass occurs when an individual element is less than 0.56 pipe diameters in length.

A comparison of three correlations for the geometric static mixer parameters of interest in this study is shown in Fig. 2. The parameters used in our calculations were: L (2 inch), D (2 inch), density (10 kg/m^3), velocity, and mixer void fraction taken from Table 2 and experimental process conditions. β was set equal to 45° , and ζ was calculated to be 0.01135 using the correlation provided by Paglianti et al. [14].

What is clear from Fig. 2 is that despite the large amount of interest in static mixer reactors with liquids/high pressure gasses as the working fluid, a robust correlation that is applicable at higher Reynolds number conditions (10,000 – 100,000) is currently lacking. Second, the role played by pipe and mixer element surface roughness characteristics has not been explored in the context of a scaling lab data to larger diameter pipes and mixers. In order to fill this void, a comprehensive experimental and simulation based study of the pressure loss behavior of a corrugated static mixer was undertaken to:

- (1) Measure and quantify the role of sand-grain surface roughness at these conditions.
- (2) Provide a more accurate, robust, and scalable pressure loss correlation.
- (3) Evaluate the efficacy of computational fluid dynamics (CFD) simulations as a pressure loss prediction and scale-up tool.

2. Methods

2.1. Measurements

2.1.1. Experimental system

To measure the experimental pressure drop across the lab scale mixing elements, two flow systems were constructed. The first system was designed to operate with compressed air up to a pressure of 120 psig. Air flow at 758.4 kPa (110 psig) was supplied by a 112 kW (150 HP) air compressor with a 3.79 m^3 (1000 US gal) holding vessel. All process piping was composed of schedule 40 stainless steel in 1-inch nominal sizing. A pipe run of 24 inches was allowed after the inlet valve to allow the flow to fully develop prior to measuring the flowrate using a cylindrical averaging differential pressure probe. The differential pressure gauges for the flowmeter and measurement section were acquired from Alicat Scientific, Inc. (Tuscon, AZ, USA) and have a calibrated accuracy of 0.2% across the entire range of Reynolds numbers. A computer data acquisition system was used to log the data from the differential pressure meters. The system was capable of producing maximum velocities of around 30 m/s in the 1-inch pipe configuration

(Re of 440,000). The piping configuration was constructed to allow the measurement section to be switched to a 2-inch nominal pipe run for measurements in a larger diameter. For all tests, a pipe run of at least 10 pipe diameters was placed before the measurement section to allow for flow development. A 1/8 inch retaining stud was fixed in the pipe to hold the mixing elements in place inside the pipe during testing.

To carry out testing in a 4-inch diameter pipe, a second experimental system was constructed to operate at near ambient pressure conditions with a 4-inch diameter PVC pipe conduit. A compressed air amplifier (Brauer Limited, Milton Keynes, UK) was used to boost compressor air to a max flowrate of 615 CFM (Re of 275,000) at near ambient conditions. The flowrate was monitored using a cylindrical averaging differential pressure probe and a differential pressure gage manufactured by Dwyer (Michigan City, IN, USA) with an accuracy of 0.5% across the measurement range.

Single corrugated mixing elements with five plates and a corrugation angle of 45° relative to the pipe axis were fabricated in for use in the 1-inch, 2-inch, and 4-inch diameter pipes. The corrugated mixer was 3D-printed from ABS material using the same CAD file utilized for the CFD simulations. A flow diagram of the first experimental set up employed to make pressure drop measurements is shown in Fig. 3. Only flow through a single mixing element with a length-to-diameter ratio of unity was investigated in this study.

In order to conduct a pressure loss test in both systems, the flowrate was incrementally increased to the maximum flowrate over a period of approximately 20 min for each test. This method allowed for enough data acquisition to construct a curve of flow versus differential pressure. Data points were collected every 230 ms via the data acquisition system. Once steady state conditions were obtained, the average differential pressure loss data was then calculated across several seconds of the steady-state condition.

Prior to carrying out the test with the single static mixer present along the flow path, the experimental configuration along with the pressure gauges were assessed for their measurement accuracies by estimating the standard error associated with the pressure drop measurements. Pressure drop data associated flow in an empty pipe was determined by gradually varying the flow rates. At each flow rate, the pressure drop data was sampled across a 20 min time period (after steady state was obtained) to obtain a standard error that was statistically invariant. The systematic errors were less than 2% above Reynolds numbers of 25,000.

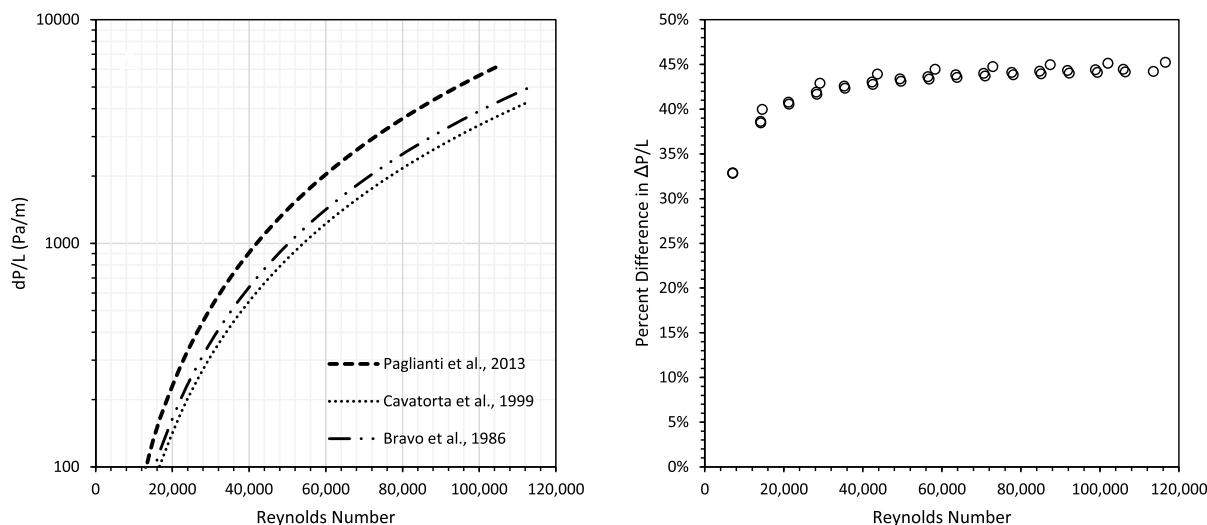


Fig. 2. Pressure drop predicted by different correlations for a single 2-inch diameter corrugated element (A) and relative difference between Paglianti et al. (2013) and Cavatorta et al. (1999) (B).

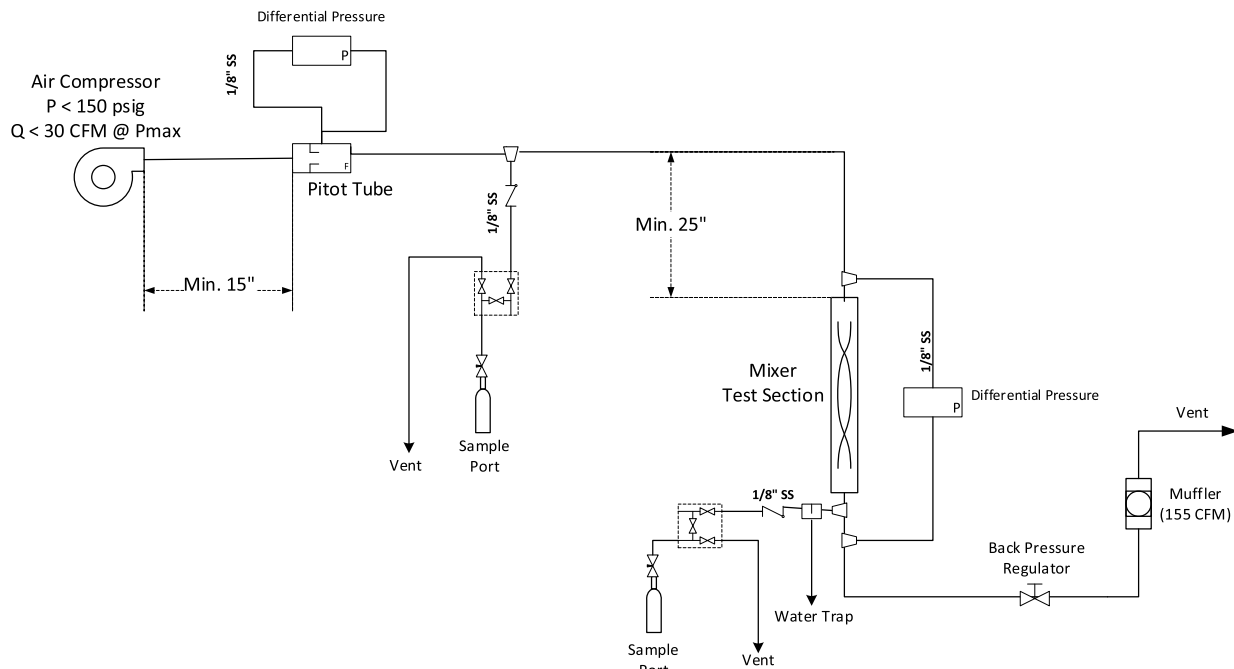


Fig. 3. Process flow diagram of first experimental system.

Table 1
Results of empty pipe experiment and fit with Colebrook-White equation.

Nominal Pipe Diameter (inch)	Reynolds Number Range	Calc'd Relative Roughness
1	15,000 - 150,000	0.17379
2	5000 - 100,000	0.11407
4	30,000 - 130,000	0.00010

2.1.2. Relative roughness estimation in an empty pipe

Since the focus of this study was to assess the pressure loss effects solely due to the corrugated static mixing elements and the pipe housing directly around the mixer, it was important to subtract the extra empty pipe pressure losses from the measurements. The pressure losses of the empty pipe housings were measured by continuously logging the flow-rate and differential pressure across the test section using a computer with data acquisition software. By fitting a line to this averaged raw data, the fanning friction factor (f) could be directly determined using Eq. (3).

The relative roughness (K_s/D) for each pipe size can be estimated from the f_D versus Re friction factor at each Reynolds number was used to fit the Colebrook-White (Eq. (5)) by iteratively solving for the relative roughness (K_s/D) by minimizing the Root Mean Square Error (RMSE).

$$\frac{1}{\sqrt{f}} = -4.0 \log_{10} \left(\frac{K_s/D}{3.7} + \frac{1.255}{Re\sqrt{f}} \right) \quad (5)$$

The result of this procedure provided the effective relative roughness for each pipe housing and is summarized in Table 1.

The baseline (empty pipe) pressure loss at each operating condition

Table 2
Experimental and CFD Model Mixer Element Dimensions.

D (in)	L/D	ϵ	τ	D_c (in)	K_s (micron)
1	1	0.756	1.32	0.126	10
2	1	0.879	1.29	0.315	10
4	1	0.879	1.29	0.693	100

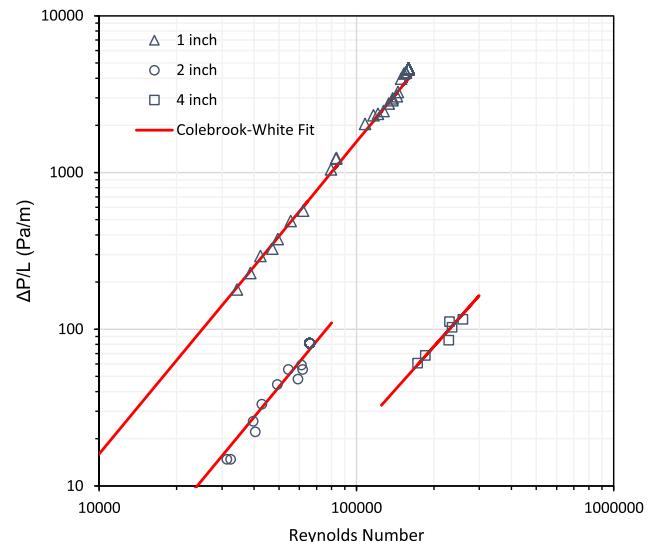


Fig. 4. Experimental pressure loss measurements and calculated pressure loss resulting from the fit of the Colebrook-White equation (Eq. (5)) for 1-inch, 2-inch, and 4-inch empty pipes.

(Reynolds number) could now be estimated by computing the friction factor using Eq. (5) and estimating ΔP using Eq. (3). A comparison of the modeled pressure loss (empirical) using Eq. (3) and actual measurements are shown in Fig. 4 to demonstrate the validity of our proposed relative roughness estimation method.

When carrying out measurements with the static mixer, this baseline pressure loss was subtracted from the measured total pressure to delineate losses attributed to the mixing element and immediate surrounding pipe housing alone. Trials were conducted at the three pipe diameters for Re ranging from 20,000 – 200,000.

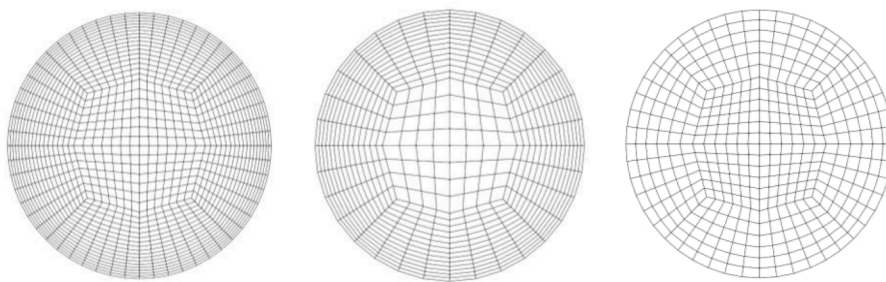


Fig. 5. Cross-sectional view of the three empty pipe meshes. From left to right: fine, medium, coarse.

2.2. Computational fluid dynamics simulations

2.2.1. Rough pipe simulations

In order to evaluate the ability of CFD simulations to account for rough pipe walls as well as determine the optimal first cell height at the pipe wall, several empty pipe meshes were constructed. Three meshes for each diameter were constructed, each with a different height of the first cell on the pipe wall. The meshes were constructed with Gmsh [16], which is an open-source meshing software capable of creating high-quality hexahedral mesh configurations with very minimal skew. Fig. 5 shows a cross-sectional view of the three meshes (fine, medium, and coarse) created for the 1-inch pipe size.

Ansys Fluent (version R2019), was evaluated for its ability to predict the empty pipe pressure loss using rough wall functions. The inlet and outlet boundaries of the 10-diameter long model pipe were considered to be periodic boundaries in order to simulate fully developed flow conditions. The SIMPLEC solver was chosen due to its proven fast, robust convergence [17]. Second-order upwind schemes were utilized for the flow and turbulence variables. Each simulation was conducted until the residuals achieved a convergence criteria of 1×10^{-5} . The near-wall region was modeled using the standard rough-wall implementation in Ansys Fluent, described in the following section. The realizable $k-\epsilon$ model was chosen for the turbulence fields due to its superior formulation for the turbulent viscosity and enhanced handling of dissipation near boundary layers [18]. At the end of the simulation, the steady-state isothermal pressure gradient was extracted. The conditions for the numerical simulations were the same as the experimental work with the working fluid modeled as an incompressible fluid with the same density of compressed air at 689.4 kPa and 21.1 °C (100 psig and 70 °F).

2.2.2. Roughness implementation in Ansys Fluent

Ansys Fluent requires the user to specify the sand-grain roughness (K_s) (estimated in Table 1) and a roughness constant (C_s). The sand-grain roughness of a surface is implemented in Ansys Fluent by shifting the law-of-the-wall mean velocity distribution by a factor (ΔB) based on the

specified surface roughness.

The shifting function (ΔB) is determined according to the boundary layer corrections provided by Cebeci and Bradshaw [19].

The hydrodynamically smooth treatment is applied when $K_s^+ < 2.25$, meaning no correction is applied to the near wall functions.

In the transitional regime ($2.25 < K_s^+ \leq 90$), Eq. (6) is applied to determine the functional correction.

$$\Delta B = \frac{1}{\kappa} \ln \left[\frac{K_s^+ - 2.25}{87.75} + C_s K_s^+ \right] \times \sin(0.4258 (\ln(K_s^+ - 0.811))) \quad (6)$$

When $K_s^+ > 90$ the shift parameter is calculated by Eq. (7).

$$\Delta B = \frac{1}{\kappa} \ln [1 + C_s K_s^+] \quad (7)$$

In Eqs. (6) and 7, the roughness constant C_s was kept at the default value of 0.5 for all simulations.

The dimensionless roughness height (K_s^+) was estimated from the sand grain roughness in Table 1 as:

$$K_s^+ = \frac{K_s C_\mu^{1/4} k^{1/2}}{\nu} \quad (8)$$

Where C_μ is an empirical model constant, k is the turbulent kinetic energy and ν is the turbulent viscosity. The definition of y^+ is given by Eq. (9).

$$y^+ = \frac{C_\mu^{1/4} k^{1/2} y}{\nu} \quad (9)$$

Once the shift parameter is determined, the non-dimensional cell wall distance y^+ can be determined by Eq. (11) in terms of the shift in the value of the dimensionless wall velocity (defined by Eq. (10)).

$$u^+ = \frac{u}{\sqrt{\frac{\tau_w}{\rho}}} \quad (10)$$

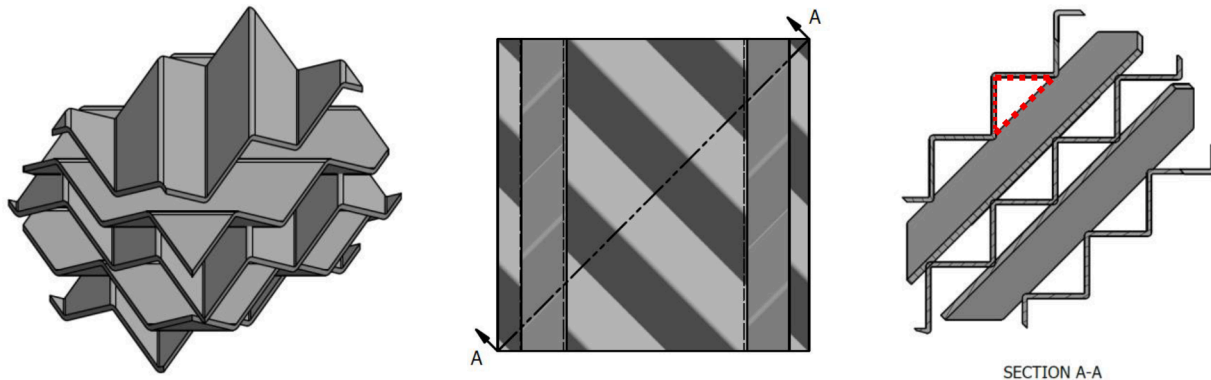


Fig. 6. 3D CAD rendering of the corrugated mixer used for CFD simulations. The red dashed lines denote the perimeter (P) of one channel used for determination of the hydraulic diameter.

Table 3
Results of grid convergence simulations.

Diameter (inch)	No. of Mesh Elements			$\Delta P/L$ (Pa/m)			% Δ , Medium-Fine
	Coarse	Medium	Fine	Coarse	Medium	Fine	
1	206,478	562,024	770,967	262,435	223,922	224,552	0.3%
2	764,284	1495,442	2570,634	15,287	14,567	13,794	-5.6%
4	1356,201	2312,566	4075,483	16,325	15,295	14,369	-6.4%

$$u^+ = \frac{1}{\kappa} \ln(Ey^+) - \Delta B \quad (11)$$

2.2.3. Mixer mesh generation and evaluation

The static mixer 3D CAD geometries were constructed from the fabricator models in order to have the most accurate similarity between the CFD and experiment. Each mixer was composed of five corrugated plates arranged at 45° with the flow axis. The mixer dimensions were such that they fit tightly into the nominal pipe size for which they were designed. Fig. 6 shows a drawing of one of the corrugated mixers from different view points.

Each mixer can be characterized by key variables such as the void fraction (ϵ), the tortuosity (τ), and the hydraulic diameter of a channel in the mixer. The void fraction is the mixer material volume divided by the total volume occupied by the mixer element. The hydraulic diameter (D_c) of the mixer channel can be estimated by calculating the diameter of an equivalent circle with the same perimeter ($\mathcal{P}$), depicted as the red dashed lines in Fig. 3) as the mixer according to Eq. (12).

$$D_c = \frac{\mathcal{P}}{\pi} \quad (12)$$

The tortuosity is defined as the length of a theoretical flow-path through the mixer divided by the actual distance traveled in the primary direction of flow [20]. Recently, Zare & Hashemabadi showed that the tortuosity could be accurately estimated by taking the average streamline length from a CFD simulation and comparing to the distance traveled in the primary flow direction [21]. Applying this same concept, the post-processed data from the simulations for all three mixer sizes was utilized to calculate the tortuosity of each mixer. A total of 10,000 streamlines were calculated based on the flow field through the mixer and this was used to extract the tortuosity for each streamline. From this, the overall average tortuosity of the mixer was calculated. Table 2 shows the key variables for each mixer element used in the experimental and CFD work.

Given the complex geometry of the corrugated mixer elements, a grid refinement study was conducted to evaluate their impact on pressure loss prediction. Three meshes of each size of mixer were created using the snappyHexMesh (a utility included in OpenFOAM ver. 1906). The mixing elements were situated in the center of the empty pipe housing mesh, which was 10 pipe diameters in length. The first mesh resolution was chosen such that the average dimensionless wall distance of the first cell (y^+) (cf. Eq. (9)) was below 30 to justify the use of enhanced wall functions. Simulations were conducted in each mesh refinement at the highest experimental Reynolds number to evaluate impact on pressure loss.

Table 3 shows the percent change in pressure loss calculated by increasing the refinement level of the mesh. Based on the results of the grid refinement study, the most refined (fine) mixer mesh was chosen at all diameters in order to minimize the impact on the simulation results.

3. Results

3.1. CFD simulations of empty pipe

The adequacy of the mesh resolution, turbulence model and sand grain roughness provided as input to Ansys Fluent was evaluated by carrying out empty pipe simulations and comparing the pressure drop

data as a function of Reynolds number. The results are shown in Fig. 7.

Good agreement is seen between the CFD predictions and measurements. The figure also shows the difference in pressure loss (i.e., $\Delta P/L$ with roughness versus $\Delta P/L$ without roughness) as a function of Reynolds number. In an empty pipe, roughness has a 70–90% contribution to the pressure loss at higher Reynolds numbers in the 1-inch and 2 inch stainless steel pipes and up to 7% contribution in the 4 inch PVC pipe that is smoother than the steel pipe (cf. Table 1).

3.2. Generalized correlation for pressure loss with static mixer

3.2.1. Pressure loss model

In order to correlate the experimental pressure loss data, the corrugated mixer was considered to be a type of porous media possessing certain geometric characteristics. The superficial velocity through the mixer can be expressed as a function of the mean flow velocity (u_0), the tortuosity (τ) and void fraction (ϵ) of the mixer [22].

$$u_s = \frac{u_0 \tau}{\epsilon} \quad (13)$$

Where $\tau = \ell/L$ and ℓ is the average streamline length through the mixer of length, L . At low Reynolds numbers, the pressure loss through a porous media is proportional to the viscous forces and the kinetic energy losses of the flow. However at high Reynolds number conditions, the kinetic-energy losses dominate flow. The capillary model presented by Comiti and Renaud is useful for understanding generalized flow through a porous media [22]. By grouping the viscous pressure loss expressed by the Poiseuille equation and the kinetic terms, they arrived at a combined pressure loss expression for flow through a packed bed [22]. In contrast to previous studies that employed the capillary model [23,24], here we choose to employ the definition of specific surface area ($a_s = 6/D_c$) for a packed bed of regular spherical particles ($\tau = 1.44$) since the tortuosity of the single corrugated mixer is close to this value [25]. This leads to the new formulation presented in Eq. (14) with the primary change being the presence of the value of 72 in the numerator as opposed to 32 in previous work [23].

$$\frac{\Delta P}{L} = \frac{72\tau^2\mu u_0}{\epsilon D_c^2} + \frac{2C_p\tau^3\rho u_0^2}{\epsilon^2 D_c} \quad (14)$$

In Eq. (14), C_p is the effective friction factor of the interior mixer channel wall. When the assumption is made that the roughness of the channels is on the same order of magnitude as the channel diameter itself, the friction factor of the channel can be estimated via the classic Nikuradse equation for fully turbulent flow in highly rough pipes in the natural logarithm form [26]:

$$\frac{1}{\sqrt{C_p/2}} = 2.46 \ln\left(\frac{D_c}{2\epsilon}\right) + 4.92 \quad (15)$$

Under the assumption that $\epsilon \cong D_c$, the friction factor for the kinetic-loss term becomes $C_p \cong 0.1936$. This assumption works well when the channels have many flow directional changes before impacting the pipe wall. However, when the channels end at the pipe wall, as in the case of the corrugated mixer in this study, a better assumption is that $\epsilon \cong D_c/2$ which leads to $C_p \cong 0.0826$ by solving Eq. (15) [22]. By re-arranging Eq. (14), a combined friction factor for the mixer as a porous media may be conveniently defined:

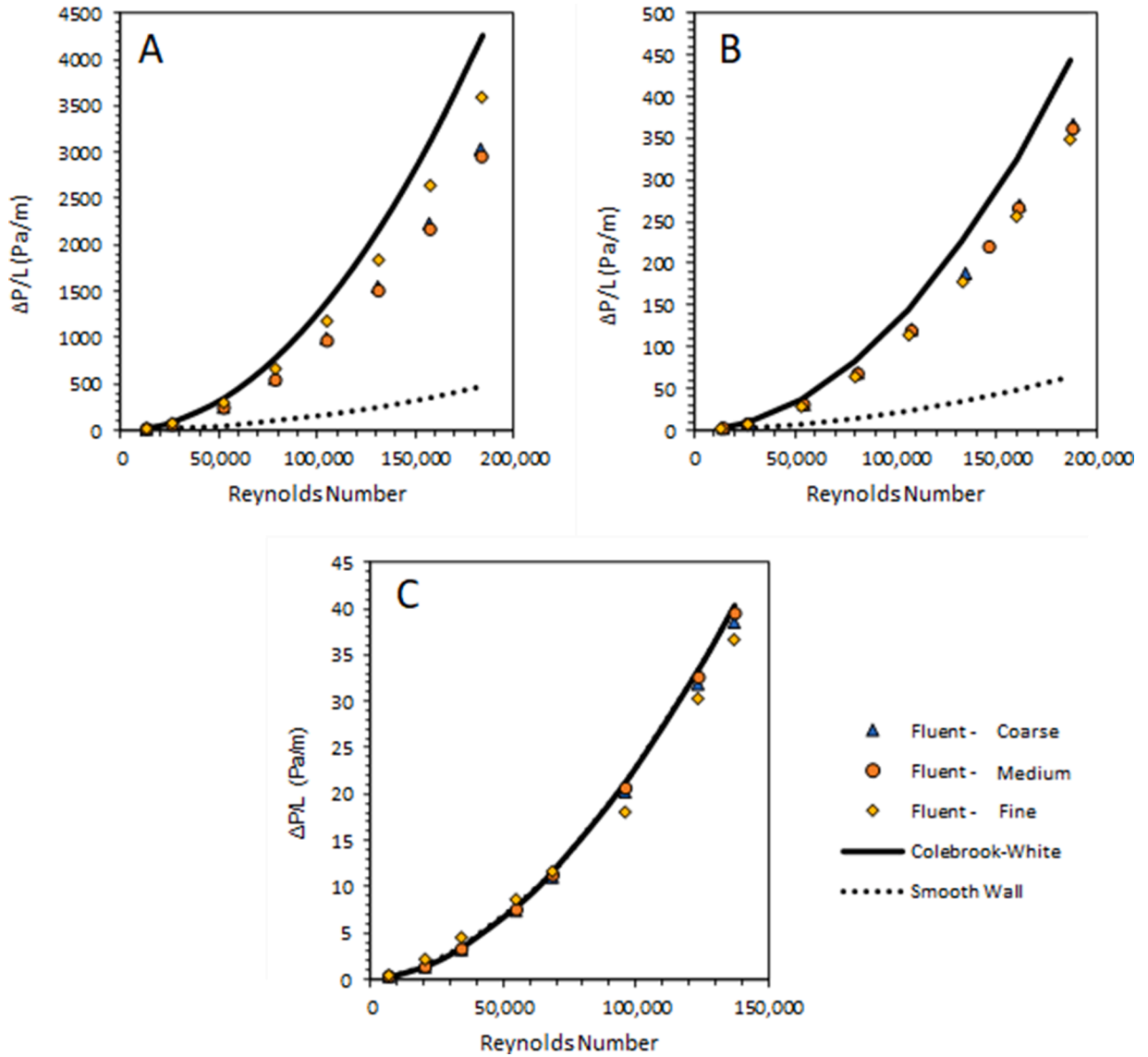


Fig. 7. Comparison of $\Delta P/L$ for an empty-pipe versus Reynolds number (fit of experimental data versus Ansys Fluent simulations at 3 mesh resolutions) for 1-inch (A), 2 inch (B), and 4 inch (C) pipe.

$$f_c = \frac{D_c \Delta P \epsilon^2}{2L \rho u_0^2 \tau^3} = \frac{36\epsilon\mu}{\rho u_0 \tau D_c} + C_p \quad (16)$$

Noting the dimensionless first term on the right hand side of Eq. (16), a Reynolds number for the mixer channel may be defined.

$$Re_c = \frac{\rho u_0 \tau D_c}{\epsilon\mu} \quad (17)$$

Finally, combining Eq. (15), 16 and 17, the semi-analytical friction factor model for the mixer in terms of the channel geometry is expressed by Eq. (18). Note the similarity in the function form of this equation and the Blasius-type equation for pressure loss (cf. Eq. (3))

$$f_c = \frac{36}{Re_c} + 0.0826 \quad (18)$$

This final expression is used as a semi-analytical basis for fitting the experimental data across the range of flow velocities and pipe diameters. Using this formulation, the pressure loss across a mixing element can be calculated by using Eq. (19).

$$\frac{\Delta P}{L} = \frac{2f_c \rho u_0^2 \tau^3}{D_c \epsilon^2} \quad (19)$$

The first term of Eq. (18) represents the Darcy contributions to flow and is primarily controlling in the low Reynolds number regimes. The second constant term is attributable to the losses associated with the rough corrugations and dominates at high Reynolds numbers. Legrande (2002) proposes that the change from Darcy flow to transitional flow occurs near $Re_c = 13$ and the fully turbulent regime starts near $Re_c = 830$.

3.2.2. Pressure loss measurements with static mixer

After collecting the pressure loss data using the laboratory systems described in Section 2, the data for one corrugated mixer for the diameters and specifications listed in Table 2 was processed to remove the losses due to the extra pipe lengths upstream and downstream of the mixer elements. Fig. 8 shows the pressure loss data (not all points shown for clarity) along with previous correlations from the literature, CFD results, and the current proposed correlation of Eq. (18) [12,14,27]. The existing correlations tended to under-predict the pressure losses

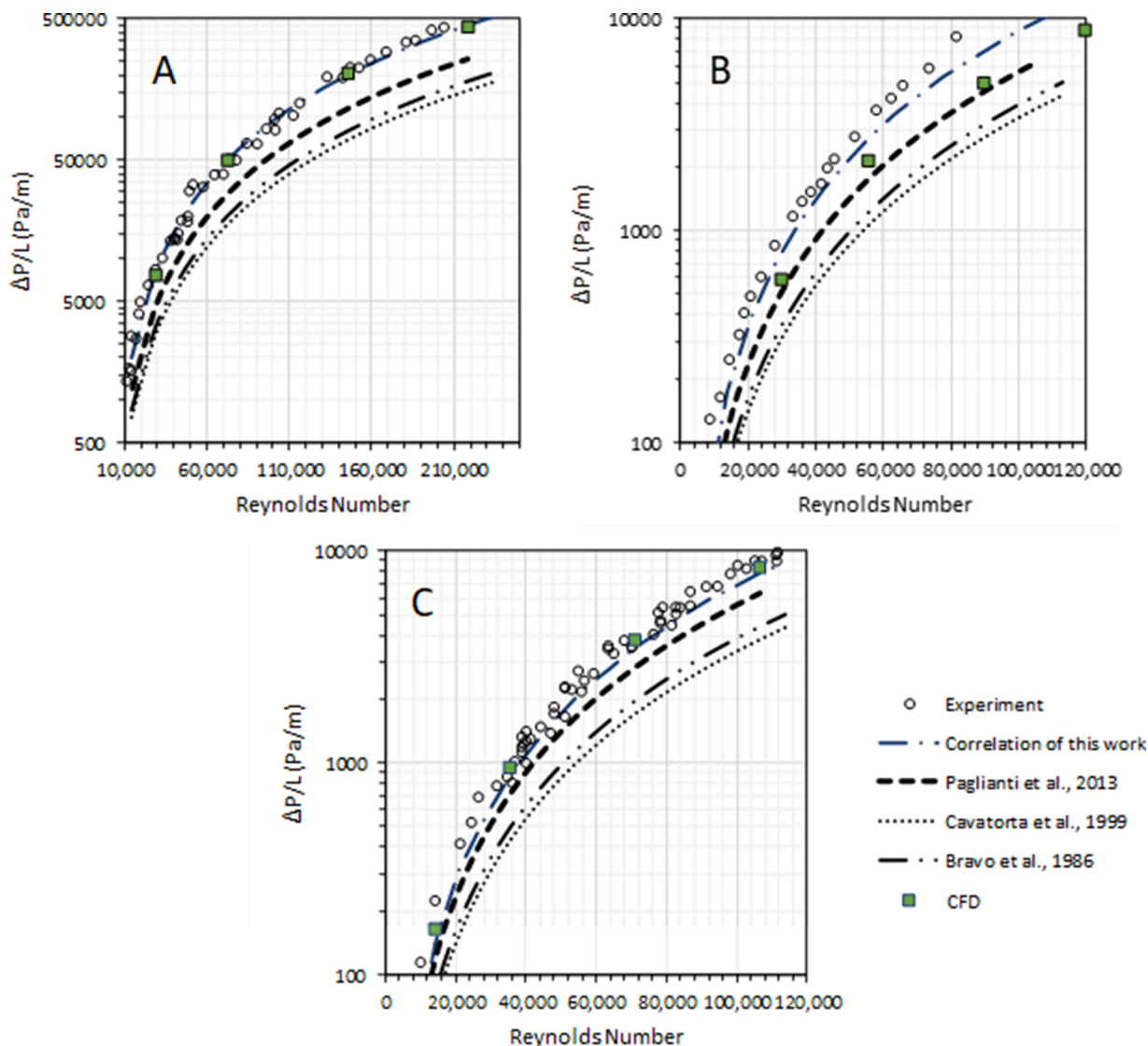


Fig. 8. Experimental data and correlation for 1-inch (A), 2-inch (B), and 4-inch (C) diameter mixer shown with literature correlations and CFD results.

compared to the experimental measurements in this study with the exception of the Paglianti et al. correlation [14] for the 4-inch diameter mixer. Overall, the proposed correlation that accounts for wall roughness effects provided better agreement with the measured data and the CFD results (carried out with wall roughness). The data for the 2-inch mixer showed more scatter and less fit compared to the other sizes. The ability of the proposed correlation to fit the data over the range of Reynolds numbers indicates that it may represent a more general formulation than previously used in the literature.

However, the CFD results obtained with Ansys Fluent were slightly lower than expected. It is suspected that this may be due to various nuisance factors that could not be accounted for in the CFD model such as irregular flows around weld locations or heterogeneous roughness in the pipe housing. In particular, the CFD simulations with the 2-inch mixer showed more deviation from the experimental results. To ascertain this, the sensitivity of the CFD simulations to wall roughness effects was carried out. Fig. 9 shows the relative error in pressure loss introduced by neglecting surface roughness as a function of Reynolds number in Ansys Fluent. In the pipe with the static mixer, roughness has up to 60% contribution to the total pressure loss at these Reynolds numbers, with higher contribution in the smaller steel pipes. This finding is critical to proper scaling of lab experiments conducted in small piping where

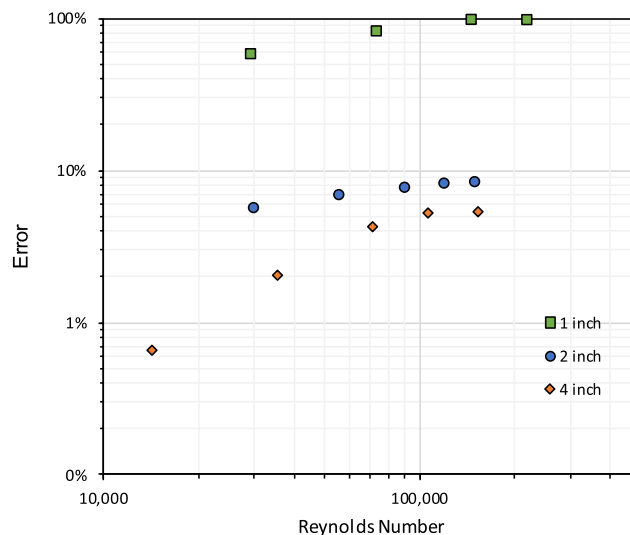


Fig. 9. Sensitivity of pressure loss calculations to surface roughness in CFD simulation for 1-inch, 2-inch, and 4-inch.

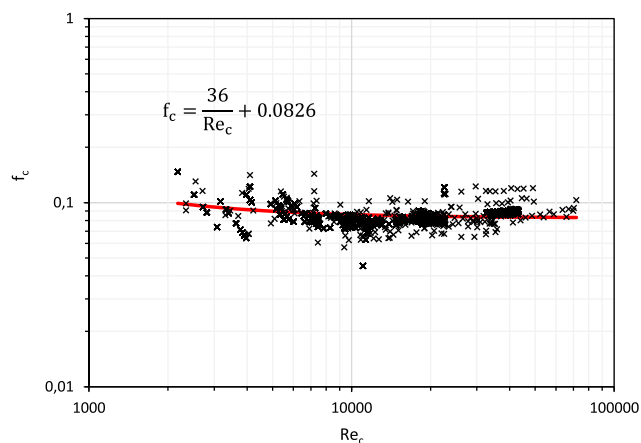


Fig. 10. Aggregated experimental data in terms of the mixer friction factor and the mixer Reynolds number.

contributions of relative roughness are significant. Direct upscaling without considering these effects at lab scales can result in incorrect predictions of pressure loss in larger pipe diameters.

3.2.3. Friction factor (f_c) estimation with static mixer

The pressure loss measurements were then employed to calculate a friction factor (f_c) based on Eq. (19). The data covered the range of $1500 < Re_c < 48,500$ which is equivalent to $8000 < Re < 2.5 \times 10^5$ in terms of the traditional Reynolds number using the pipe diameter as the characteristic dimension. As shown in Fig. 10, the data collapsed very well into the semi-analytical model of Eq. (18). The model (cf. Eq. (18)) exhibited a mean absolute percentage error (MAPE) of 19.5% with a total of 6160 data points collected during this study. Additionally, the model (cf. Eq. (18)) had MAPE of 15.5% when compared to low Reynolds number data collected from the literature as shown in Fig. 11. This suggests that this formulation provides a reasonable approximation for the pressure losses through the corrugated geometry.

The mixer geometrical details given in each reference were used for estimating the geometric parameters (L , D) and the tortuosity was estimated from the values calculated using CFD in this study while interpolating for the changes due to differences in diameter and void fraction. The data extracted from the literature was fit surprisingly well by the correlation, even considering the disparate experimental conditions and mixer geometrical details. The advantage of the proposed model (cf. Eq. (18)) over previous models lies in the fact that all the data extracted from the literature and measured in this study is well fit by the single model. Additionally, all key geometrical details governing the flow are taken into account by including the tortuosity and void fraction of the mixer in the calculations, as opposed to only the void fraction as has been done in previous studies. Inclusion of the tortuosity allows the data to collapse to a single correlation that accounts for differences in plate thickness, mixer fit, and length. It is possible that the tortuosity could be eliminated from the model by expressing as a function of the void fraction, similar to previous work in the area of porous media research [20]. This modification would simplify the model for ease of use, however this would require additional study.

Previous work has defined the flow regime boundaries as the point at which the contributions of the viscous or inertial forces was less than 5% for the turbulent regime and the Darcy regime, respectively [28]. Adopting this criteria, the critical channel Reynolds number at which turbulent flow starts is $Re_{cT} = 8,300$ which is somewhat higher than the values reported in previous works. Lage & Antohe cite $Re_{cT} = 1,570$ as the transition to turbulent dominated flow [28]. Similarly, other work uses $Re_{cT} = 830$ as a criteria for this transition [29].

Finally, the Z-factors for the single mixer element were estimated (cf. Eq. (1)) from the experimental data in the different pipe sizes as a

function of Reynolds number. Fig. 12 shows that the Z-factor ranges from 80 – 400 and is highly dependent on pipe diameter and roughness. This may indicate that the Z-factor is not a completely satisfactory method for pressure loss determination.

4. Conclusion

Robust, scalable correlations for pressure loss across SMV static mixers at high Reynolds (Re) numbers are currently lacking. This is first demonstrated by showing up to 45% deviation among current correlations at $Re > 10,000$. To fill this void, an experimental and computational fluid dynamics (CFD) based study of a single corrugated static mixer for Reynolds numbers in the range of 10,000 – 250,000 is reported. Measurements of pressure loss were carried out in pipes of diameters of 1-inch, 2-inch, and 4-inch with significant variations in the pipe relative roughness.

The equivalent sand grain roughness was first estimated from the pressure loss measurements in an empty pipe and was provided as an input to the CFD model. The agreement of the CFD predictions with the measurements ascertained the validity of the roughness estimate. In an empty pipe, roughness contributed about 70–90% of the pressure loss at higher Reynolds numbers in the 1-inch and 2-inch stainless steel pipes and up to 7% of pressure loss in the 4 inch PVC pipe.

A new semi-analytical correlation for pressure loss across an SMV static mixer at high Reynolds (Re) numbers was proposed. The measured pressure loss data was compared to the new proposed correlation as well as existing correlations from the literature. It was revealed that the existing correlations tend to under-predict the pressure losses compared to the measured data. On the other hand, our proposed correlation showed an improved agreement with the pressure loss data generated in this study and also provided a satisfactory representation of the applicable raw data from the literature. The new correlation was also deemed to be more broadly applicable to a variety of static mixer configurations since it includes the effects of key geometric parameters such as tortuosity and void fraction. Additionally, the semi-analytical approach includes the impact of hydrodynamic roughness induced by the corrugated plate shape. CFD simulation was used to facilitate scale-up and delineate the pressure losses arising from roughness and manufacturing defects in these configurations. With the static mixer, roughness had up to a 60% contribution to the overall pressure loss, with the highest contributions in the 1-inch pipe.

Overall, this study was able to provide a more robust correlation for the pressure loss in corrugated static mixers that may be easily extensible to geometrical variants without the need for additional measurements. Additionally, well resolved CFD simulations that include the effects of wall roughness were shown to be an acceptable tool for

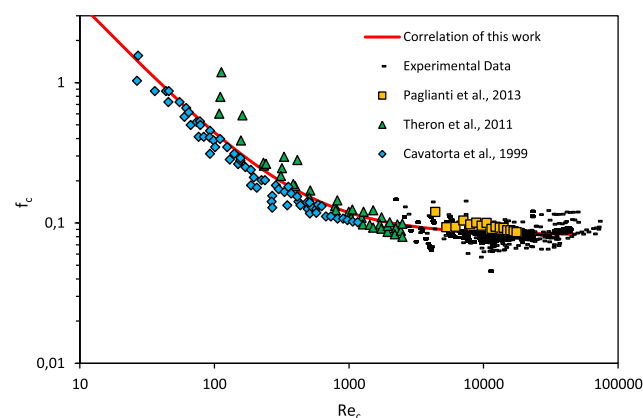


Fig. 11. Literature data for single mixer-element pressure loss compared to the correlation of Eq. (18) in terms of the mixer friction factor and the mixer Reynolds number.

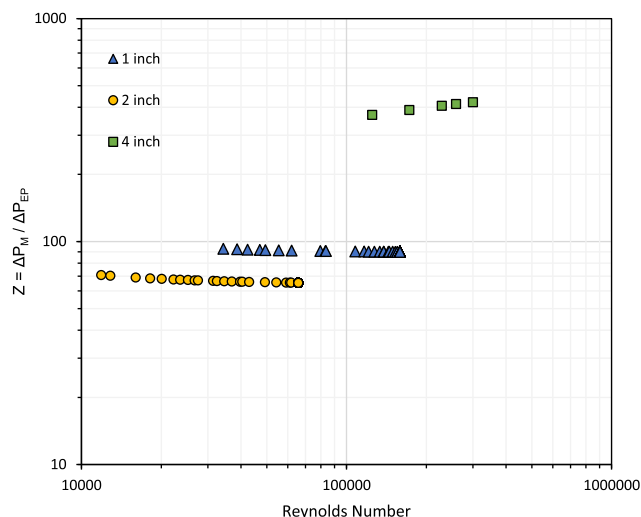


Fig. 12. Z-factor estimates calculated from the experimental data for mixer and empty pipe (not all data points shown for clarity).

evaluating the pressure losses which further validates this method for scale-up predictions when data at larger scales are not available. Future work will examine how the proposed correlation and CFD simulation can be used to predict pressure losses in different linear mixer configurations and orientations as well as developing links to mass transfer behavior in these types of mixers.

Author contribution statement

E. Lowry and Y. Yuan conceived of the presented idea and designed the study with support from G. Krishnamoorthy. E. Lowry developed the theory and performed the experimental measurements and CFD computations. Y. Yuan verified the experimental and data analytical methods. G. Krishnamoorthy provided guidance and oversight for all the CFD simulation work. All authors discussed the results and contributed to the final manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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